

Multiple Choice

1. **Answer:** D

Options: A) electromagnetic radiation.; B) thermal conduction.; C) gravitational pressure.; D) radioactivity.; E) nuclear fusion.

Note: Correct option: D - radioactivity.

2. **Answer:** B

Options: A) Gluon; B) Electron; C) Quark; D) Proton; E) Photon

Note: Correct option: B - Electron

3. **Answer:** A

Options: A) 7.21×10^{25} nuclei; B) 9.09×10^{-26} nuclei; C) 4.48×10^{-26} nuclei; D) 26.98×10^{-27} nuclei; E) nuclei

Note: Correct option: A - 7.21×10^{25} nuclei

4. **Answer:** A

Options: A) 0.75; B) 0.40; C) 0.55; D) 0.65; E) 0.50

Note: Correct option: A - 0.75

5. **Answer:** A

Options: A) From accretion within the solar nebula at the same time the planets formed; B) From fragments of planets ejected by impacts; C) From dust grains that escape from passing comets; D) From the remnants of a planet's atmosphere; E) From the gravitational pull of nearby planets

Note: Correct option: A - From accretion within the solar nebula at the same time the planets formed

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '880 lb'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '72800 volts and 0.21 A'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' 1.74 sec^{-2} '.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Increases by a factor of 81.'

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 2 W. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 4.2. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key